

# PRASHANT AGRAWAL

SOFTWARE ENGINEER | PYTHON | LANGGRAPH | RAG & AGENTIC AI | FASTAPI | FULL-STACK DEVELOPMENT

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## PROFESSIONAL SUMMARY

Software Engineer with hands-on experience building agentic AI systems, RAG pipelines, LangGraph-generated data-transformation workflows, and full-stack backend platforms. Skilled in LangChain/LangGraph-based agent orchestration, Natural Language-to-SQL pipelines, FastAPI backend development, and data security/compliance workflows. Strong foundation in full-stack development (MERN) and database design with PostgreSQL and MongoDB, with working knowledge of Docker, JWT authentication, and cloud-integrated AI tooling.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, C++, SQL

**AI / Agentic Systems:** LangChain, LangGraph, RAG Pipelines, NL-to-SQL, Agentic Systems, Data Masking (HIPAA)

**Backend & Databases:** FastAPI, Node.js/Express (MERN), React, REST APIs, JWT Auth, PostgreSQL, MongoDB

**Cloud, DevOps & Tools:** Docker, Caching, Git, Deployment

## PROFESSIONAL EXPERIENCE

### Argusoft Pvt Ltd

Gandhinagar

#### Programmer Analyst

Jan 2026 - Present

- Built a HIPAA-compliant AI platform, implementing automated PII/PHI data masking to secure sensitive healthcare data across agentic workflows.
- Designed a truly agentic AI agent architecture capable of autonomous multi-step reasoning and tool use within the compliance platform, improving response reliability by 50%.
- Developed an end-to-end RAG and Natural Language-to-SQL (NL-SQL) pipeline, enabling secure, compliant natural-language querying over sensitive structured data.
- Designed and implemented a LangGraph-based workflow-generation engine that assembles data-transformation pipelines from natural-language queries and target output formats, reducing workflow-building time by 90%.
- Built an edit/iteration capability for existing generated workflows and introduced caching, reducing compute cost by 10%, with edits recomputing only affected steps to cut turnaround time by 25%.
- Performed extensive testing and validation of AI agents across several projects, executing diverse queries and identifying edge cases, hardening reliability by 10%.

#### Programmer Analyst Trainee

Jul 2025 – Dec 2025

- Gained hands-on experience in FastAPI, building scalable backend systems with authentication, role-based access control, and optimized APIs.
- Explored Deep Learning and Generative AI, including working with RAG pipelines, LangChain, and LangGraph for building intelligent applications.
- Designed and developed a full-stack E-Learning platform with features like course/module management, quiz system with automated scoring, enrollments, and progress tracking.
- Planned and structured a LangGraph-based workflow for an existing LangChain project, improving orchestration and modular flow design.

#### Programmer Analyst Intern

Feb 2025 – Apr 2025

- Worked on Checkmate, a checklist and workflow automation platform, focusing on backend development with Java Spring Boot — designing RESTful APIs for checklist creation, task assignment, and role-based workflow execution, and modeling data (JPA/PostgreSQL) for tasks, teams, reminders, dependencies, and audit logging.

## PROJECTS

### Real Estate App | MERN Stack, JWT Authentication

- Built a MERN-based real estate application enabling users to securely list, browse, and connect around property listings, with JWT authentication for safe account management.

### Blog App | MERN Stack

- Developed a MERN-based blogging platform allowing users to create and manage posts on a secure platform with public read access and protected content management.

## EDUCATION

**B.Tech, Computer Science and Engineering — G.L. Bajaj Institute of Technology and Management, Greater Noida** 2021 - 2025

CGPA: 8.1/10

Relevant Coursework: Object Oriented Programming, Databases, Discrete Maths, Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning, Advanced Data Structures and Algorithms, Web Designing